

Diffusion Language Models

Recent Developments and the Core Accuracy Gap with Autoregressive Models

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Abstract

Diffusion language models (DLMs) generate text by iteratively denoising a fully-masked (or noised) sequence rather than predicting one token at a time left-to-right. They promise *parallel*, any-order generation, native infilling, and bidirectional context. Over 2024–2026 the approach matured from a research curiosity into billion-parameter systems (LLaDA, Dream, Mercury, Gemini Diffusion) that approach autoregressive (AR) quality on many tasks. Yet a persistent quality gap remains. This report surveys the recent landscape and then focuses on the question of *why*: the fundamental accuracy limitation is the combination of a **looser variational training objective** and a **conditional-independence approximation made during parallel decoding**, which together create an unavoidable quality–parallelism trade-off.

1 Background: how diffusion language models work

An AR model factorizes the joint distribution exactly via the chain rule, $p(x_{1:L}) = \prod_i p(x_i | x_{<i})$, and is trained by exact maximum likelihood (teacher forcing). A diffusion LM instead defines a *forward corruption process* that gradually destroys a clean sequence x_0 into noise x_T , and learns to *reverse* it. Two families exist:

- **Continuous diffusion** (e.g. Diffusion-LM, 2022) embeds tokens in a continuous space and runs Gaussian diffusion, rounding back to tokens. It struggles because text is fundamentally discrete; rounding is lossy.
- **Discrete diffusion** corrupts tokens directly. The dominant variant is **masked (absorbing-state) diffusion**: the forward process replaces tokens with a special [MASK] at a time-dependent rate, and the model learns to predict the original tokens at masked positions — a time-conditioned, generative generalization of BERT-style masked prediction.

Masked diffusion now dominates because of cleaner objectives and strong empirical results. Key methodological steps were D3PM (discrete diffusion framework, 2021), SEDD (score-entropy discrete diffusion, 2023), and the simplified masked objectives of MDLM / MD4 (2024), which showed the training loss reduces to a clean weighted average of masked-token cross-entropies.

2 Recent developments (2024–2026)

2.1 Scaling to billions of parameters

LLaDA (Large Language Diffusion with mAsking, 2025) trained an 8B masked diffusion model from scratch and showed it competitive with LLaMA-class AR models of similar scale on standard benchmarks and in-context learning — the clearest evidence that the paradigm scales. **Dream 7B**

(2025) reached similar quality with a strong open release. These results established that DLMs are not inherently capped at small scale.

2.2 Commercial, speed-oriented systems

Mercury (Inception Labs, 2025) and **Gemini Diffusion** (Google, 2025) are production diffusion LLMs marketed primarily on *throughput*: because tokens are produced in parallel, they report very high tokens/second, especially for code. These systems are the practical argument for the paradigm — when the parallelism can be realized, latency drops sharply.

2.3 Bridging AR and diffusion

Block diffusion (BD3-LMs, 2025) partitions the sequence into blocks, running AR *across* blocks and diffusion *within* each block. This interpolates between the two regimes, recovers the ability to generate variable-length sequences and use a KV cache, and improves likelihood — a direct response to the gap discussed below. “Semi-autoregressive” decoding more broadly is now a common quality lever.

2.4 Other threads

Adaptive/confidence-based **remasking** and planner-guided **decoding order**; **distillation** and consistency-style few-step samplers to cut the number of denoising iterations; **unified multimodal** diffusion (text + image) models; and early work on diffusion models for **reasoning**, where parallel revision is a natural fit but verifiable accuracy still trails strong AR reasoners.

3 The fundamental accuracy problem vs. autoregressive models

The recurring observation is that, *at matched parameters and training compute*, masked diffusion LMs reach somewhat **worse likelihood/perplexity** than AR models, and that closing the gap tends to **erase their speed advantage**. This is not an incidental engineering issue; it has two coupled roots.

3.1 Root 1 — A looser training objective (the likelihood gap)

AR training optimizes the *exact* log-likelihood, because the chain-rule factorization is exact. Diffusion training instead optimizes an **evidence lower bound (ELBO)** — a variational *upper bound* on the negative log-likelihood. For masked diffusion the bound takes the clean form

$$-\log p_{\theta}(x_0) \leq \mathbb{E}_{t \sim U(0,1)} \mathbb{E}_{x_t} \left[\frac{1}{t} \sum_{i \in \mathcal{M}(x_t)} -\log p_{\theta}(x_0^{(i)} \mid x_t) \right], \quad (1)$$

i.e. a time-weighted cross-entropy over *only the masked positions* $\mathcal{M}(x_t)$. Two consequences:

- **The bound is loose.** Optimizing an upper bound on NLL does not optimize NLL directly; there is slack between the ELBO and the true likelihood that AR models simply do not pay.
- **Weaker supervision per example.** At a given noise level only the masked tokens contribute to the loss, so each training sequence yields signal on a *fraction* of its positions (averaged over noise levels, far less than AR’s “every token, every step”). This lowers data/compute efficiency — the model effectively learns from fewer prediction targets per token of data.

3.2 Root 2 — The conditional-independence approximation at decoding

This is the deeper, more fundamental issue. Suppose the current partially-masked sequence is x_t and the model unmask a set S of positions in a single denoising step. The model provides a *marginal* distribution $p_\theta(x_0^{(i)} \mid x_t)$ for each position $i \in S$, and parallel sampling draws them **independently**:

$$p_\theta(x_S \mid x_t) \approx \prod_{i \in S} p_\theta(x_0^{(i)} \mid x_t). \quad (2)$$

But the *true* joint over those positions does not factorize — simultaneously-generated tokens are correlated (think of choosing two words of a phrase that must agree). Equation (2) replaces the joint with a **product of marginals**, discarding all within-step dependencies. The error this introduces is exactly the mutual information among the tokens generated together.

The trade-off this creates is the crux:

- Unmask **many** tokens per step $\Rightarrow$ few steps, high parallelism, *large* independence error $\Rightarrow$ degraded coherence and likelihood.
- Unmask **one** token per step $\Rightarrow$ the conditional is exact (no independence assumption), recovering AR-quality *but with AR-like cost* (L sequential steps) — and indeed single-token-per-step masked diffusion is essentially any-order autoregression.

So the very mechanism that gives DLMs their speed (parallel, multi-token unmasking) is the mechanism that costs them accuracy. **You cannot have full parallelism and exact joint modeling at the same time.**

3.3 Why AR avoids both problems

AR pays neither cost: its factorization is exact (no ELBO slack), and it never samples two tokens “simultaneously” — each token is conditioned on all previously sampled tokens, so all dependencies are captured by construction. The price AR pays is the opposite one: strictly sequential generation (no intra-sequence parallelism) and unidirectional context.

3.4 How the field is attacking the gap

Every major quality lever targets Root 2 by *shrinking the set unmasked per step* or *re-introducing dependencies*: semi-autoregressive / block diffusion (AR across blocks limits the independence error to within a block); confidence-based decoding that unmask only high-certainty (low-correlation) tokens first; predictor–corrector and remasking samplers that revisit earlier choices; and distillation that tries to push more joint structure into fewer steps. Each narrows the gap but trades back some of the parallelism that motivated diffusion in the first place.

4 Summary

Bottom line. Diffusion LMs have become genuinely competitive at scale and can be dramatically faster when their parallelism is realized. Their fundamental accuracy limitation against AR models is structural, not a tuning problem: training optimizes a *looser variational bound* with weaker per-token supervision, and fast parallel decoding relies on a *conditional-independence approximation* that ignores correlations among tokens generated in the same step. Matching AR quality requires taking more denoising steps (unmasking fewer tokens at a time), which converges

	Autoregressive	Masked diffusion
Factorization	Exact chain rule	Variational (ELBO upper bound)
Training signal	Every token, exact MLE	Masked tokens only; looser bound
Generation	Sequential, 1 token/step	Parallel, many tokens/step
Key approximation	none (exact)	tokens in a step drawn independently
Strength	accuracy, likelihood	latency / throughput
Core weakness	no intra-sequence parallelism	quality drops as steps decrease

Table 1: The trade-off in one table.

toward autoregression and surrenders the speed advantage. The most promising direction — block / semi-autoregressive hybrids — explicitly navigates this trade-off rather than escaping it.

Selected references. Austin et al., *Structured Denoising Diffusion (D3PM)*, 2021; Li et al., *Diffusion-LM*, 2022; Lou et al., *Discrete Diffusion with Score Entropy (SEDD)*, 2023; Sahoo et al., *Simple and Effective Masked Diffusion (MDLM)*, 2024; Shi et al., *Simplified Masked Diffusion (MD4)*, 2024; Nie et al., *LLaDA: Large Language Diffusion Models*, 2025; Arriola et al., *Block Diffusion (BD3-LMs)*, 2025; Inception Labs, *Mercury*, 2025; Google DeepMind, *Gemini Diffusion*, 2025.